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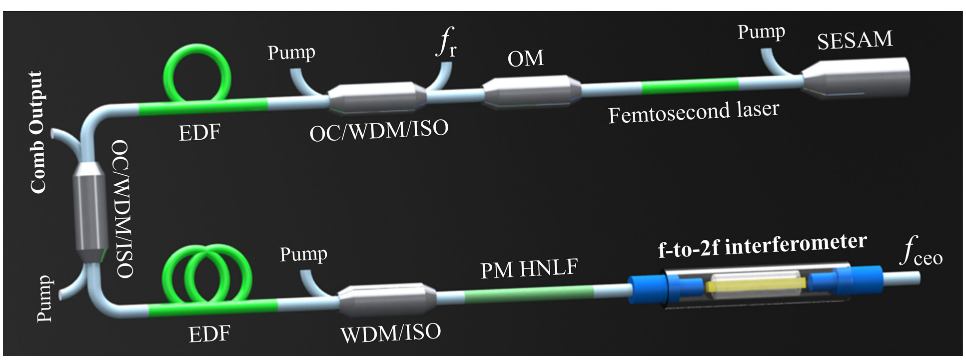

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Compact, High-Performance All-Polarization-Maintaining Er: Fiber Frequency Comb With Single Fiber Actuator

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Abstract: The large volume and weak environmental adaptability of fiber optical frequency combs (OFCs) have become the main obstacles for their applications in various fields. To address these issues, in this study, we present a compact, low-cost f-to-$2f$ interferometer and fiber actuator with a large tuning range and a high control bandwidth for a 200-MHz OFC that is based on a 1.5-μm all-polarization-maintaining fiber mode-locked laser. By employing customized fiber-coupled gradient index lenses, our f-to-$2f$ interferometer is encapsulated in a miniature tube with a diameter of only 4 mm and a length of 40 mm, which substantially reduces the optical section size of the frequency comb as compared to conventional devices. The carrier envelope offset beat with a signal-to-noise ratio of 40 dB is detected in a resolution bandwidth of 360 kHz. In addition, a laboratory-made piezoelectric transducer-driven mechanical actuator for repetition rate regulation exhibited a large tuning range of 106 kHz (corresponding to an effective temperature drift of 53 °C) and a high control bandwidth of approximately 1 kHz. This resulted in a robust repetition rate locking with an Allan deviation of 330 μHz at a gate time of 1 s and a residual integrated timing jitter of 418 fs [3 Hz to 1 MHz] when referenced to a hydrogen maser. Along with reducing the size and improving the environmental adaptability of the OFC, our design can also decrease the power consumption of the system significantly. Our findings provide a new direction to the development of OFCs for various applications.

Index Terms: Polarization-maintaining fiber, frequency comb, fiber actuator, control bandwidth.

1. Introduction

The advent of the optical frequency comb (OFC) bring a revolutionary breakthrough in precision optical frequency metrology by providing a direct and simple link between optical oscillation and microwave frequencies [1], [2]. Their ability to convert between optical and microwave frequencies as well as their broad spectral bandwidth [3] and high accuracy characteristics [4] make OFCs powerful tools in the fields of low-phase-noise microwave generation [5], [6], comb-based spectroscopy [7], [8], high-precision laser detection and ranging [9], [10], and atomic clock networks [11], [12]. With the rapid growth in these applications, portable, robust, low-power-consumption, and inexpensive OFCs are increasingly necessary. Although chip-level frequency comb sources based on microresonators and semiconductor lasers have shown great potential to become practical future alternatives, they still face many challenges, such as low optical efficiency, large propagation loss, and low photonic integration [13]. Thus, there is still a requirement for research on these devices before they can be extensively applied. At present, from the perspective of cost, robustness, and complexity, the more attractive solution for fieldable combs is fiber-based OFCs built with all-polarization-maintaining (PM) fiber and off-the-shelf fiber components [14]–[16]. With the transition of fiber OFC, especially OFCs based on erbium-doped fiber laser, from laboratory research to outdoor applications, in different outdoor environments, including moving vehicles and sounding rockets, highly reliable and high-performance fiber OFCs have been reported [17]–[19]. Further reducing the volume of fiber OFCs, decreasing the power consumption, and improving their environmental adaptability has become an important research direction.

For a typical self-referenced all-fiber frequency comb, the collinear f-to-$2f$ interferometer used to detect the carrier envelope offset (CEO) signal usually comprises four aspheric lenses for laser collimation and focusing and a periodically poled lithium niobite (PPLN) crystal for frequency-doubling [20]. This bulky interferometer occupies almost half of the volume of the optical section. Additionally, counteracting the repetition rate (f_{r}) drift caused by ambient temperature changes usually requires an active temperature control with a high dynamic range for the mode-locked laser. This temperature control unit typically consists of a thermal-electric cooler (TEC), heat sink, temperature sensor, and temperature feedback control circuit. Invariably, these components increase the size and power consumption of the OFC and result to a poor environmental adaptability of the OFC as well. To address these issues, researches are actively underway in the field of OFC design. Some groups have employed a commercial, fiber-coupled, ridged, waveguide PPLN-based f-to-$2f$ interferometer to detect the CEO beat signal, which considerably simplifies and improves the CEO beat detection [15], [21], [22]. However, the fiber lead on this fiber-coupled waveguide PPLN is long ($>$10 cm), complicating the ability to compensate for group velocity walk-off between the fundamental and doubled light pulses by adding an "in-line" PM fiber interferometer. Furthermore, the waveguide PPLN is expensive. On the other hand, different actuators with large tuning ranges have been designed and used to control the cavity length. Manurkar *et al.* realized a "fiber resistive modulator" with an f_{r} tuning range of more than 4.2 kHz (corresponding to an effective temperature change of 4.2 °C) and a response time of 0.5 s by resistive heating of the intracavity gold-palladium-coated fiber [23]. Optical delay lines or other translating optics can provide a large f_{r} tuning range of over 1 MHz [24], [25], however, it is quite slow and also not suitable for high-repetition-rate fiber lasers. Naturally, a fiber actuator with both high control bandwidth and large tuning range is desired.

In this study, we demonstrated a compact, fully optically integrated all-PM fiber OFC based on a 1.5-μm erbium-doped fiber oscillator. A fiber-coupled, fully optically integrated, miniature f-to-$2f$ interferometer based on gradient index (GRIN) lenses is presented. The entire interferometer is sealed in a micro-plastic tube with a diameter of only 4 mm and a length of 20 mm. The CEO beat with a signal-to-noise ratio (SNR) of 40 dB is detected by using this miniature f-to-$2f$ interferometer. Compared to the previously mentioned bulky f-to-$2f$ interferometer setups and waveguide designs, our design is compact and inexpensive. Furthermore, we realized a piezoelectric transducer (PZT)-based mechanical actuator for f_{r} regulation with a tuning range of 106 kHz and control bandwidth of approximately 1 kHz. The large-tuning-range actuator makes it possible to achieve long-term f_{r} locking even without active temperature control for the oscillator. This design not only reduces the size and power consumption of the OFC, but also can significantly improve its robustness and environmental adaptability. In addition, we determined the detailed static and dynamic response of f_{r} for the laboratory-made fiber actuator modulation. Finally, we characterized the in-loop frequency

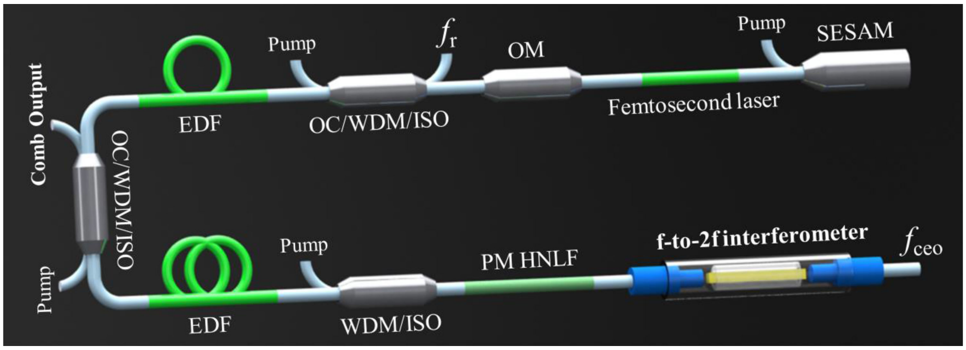

Fig. 1. Detailed schematic of the compact, all-PM-fiber OFC. SESAM: semiconductor saturable absorber mirror, EDF: erbium-doped fiber, OM: output mirror, WDM: wavelength-division multiplexer, ISO: isolator, OC: output coupler, and HNLF: highly nonlinear fiber.

stability of f_r and f_{ceo} by referencing a hydrogen maser, and also demonstrated the noise property of f_r in the free-running and phase-locked state.

2. Optical System Design for Frequency Comb

The basic design of the compact, optically integrated frequency comb is presented in Fig. 1. It is notable that the OFC is completely based on PM fiber and off-the-shelf PM optical components. In addition, there are no free-space parts, which makes the OFC more compact and robust. The oscillator design is similar to that reported in Ref. [26]. The cavity length of the oscillator is approximately 50 cm, which corresponds to a repetition rate of approximately 200 MHz. The mode-locked laser emits 230-fs pulses at a center wavelength of 1560 nm with a spectral bandwidth of 11 nm and an average output power of 3.7 mW. The output pulse energy is amplified to 1.7 nJ (with a spectral bandwidth of 33.14 nm, as shown in Fig. 3(a)) by a two-stage PM erbium-doped fiber amplifier (EDFA). The first stage provides the functions of pulse stretching and energy preamplification. The second stage is designed to boost the pulse energy and further broaden the spectral width. Here, a 2.8-m-long low-gain PM erbium-doped fiber ($D = -27.99$ ps/nm/km, peak absorption rate (PAR) $= 25$ dB/m at 1530 nm) in the first stage and a 1-m-long high-gain PM erbium-doped fiber ($D = -12$ to -18 ps/nm/km, PAR $= 98$ dB/m at 1530 nm) in the second stage are used as the gain medium. Laser pulses from the oscillator and the first amplification stage are coupled into the next stage through a compact combined component (OC/WDM/ISO) that includes an output coupler (90:10), a wavelength division multiplexer (WDM), and a PM isolator. Then, 10% tap output power (approximately 300 μW) from the first OC/WDM/ISO is used for the generation and locking of f_r. Three 976-nm pump laser diodes with a total pump power of 1400 mW are used to achieve laser pulse energy amplification and spectrum broadening.

In order to detect the carrier envelope offset frequency (f_{ceo}), the amplified laser pulses are compressed to 63 fs (see Fig. 3(b)) by a 130-cm standard single-mode PM 1550 fiber before propagating them into a 62.5-cm PM highly nonlinear fiber (HNLF). The generated supercontinuum (SC) spectrum spans a full octave ranging from approximately 1015 nm to approximately 2030 nm (Fig. 3(c)). This spectrum is then transmitted into the fiber-coupled, GRIN lens-based micro f-to-2f interferometer that doubles the light frequency at 2030 nm and compares it to the fundamental light at 1015 nm. The design of the miniature f-to-2f interferometer is illustrated in Fig. 2. This interferometer is enclosed in a plastic tube with a diameter of 4 mm and a length of 20 mm. It consists of a 10-mm-long MgO:PPLN crystal for frequency-doubling, two pigtailed GRIN lenses, and two plastic tubes for encapsulating the entire interferometer. On the input side, the pigtailed PM 1550 fiber of the GRIN lens is used to compensate for the group-velocity walk-off between the fundamental light and doubled-frequency light component. There is only a 6-mm-long fiber lead

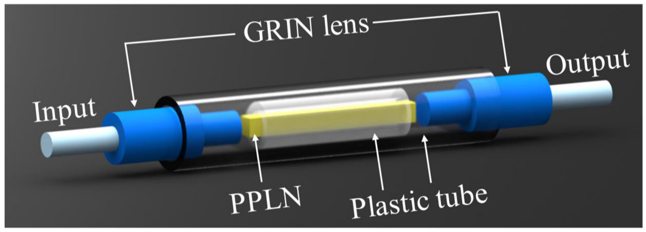

Fig. 2. Schematic of the integrated miniature f-to-2f interferometer containing two single-mode PM 1550 fiber-coupled GRIN lenses as the input and output ports. All components of the miniature interferometer are bonded by a UV glue with a low expansion coefficient.

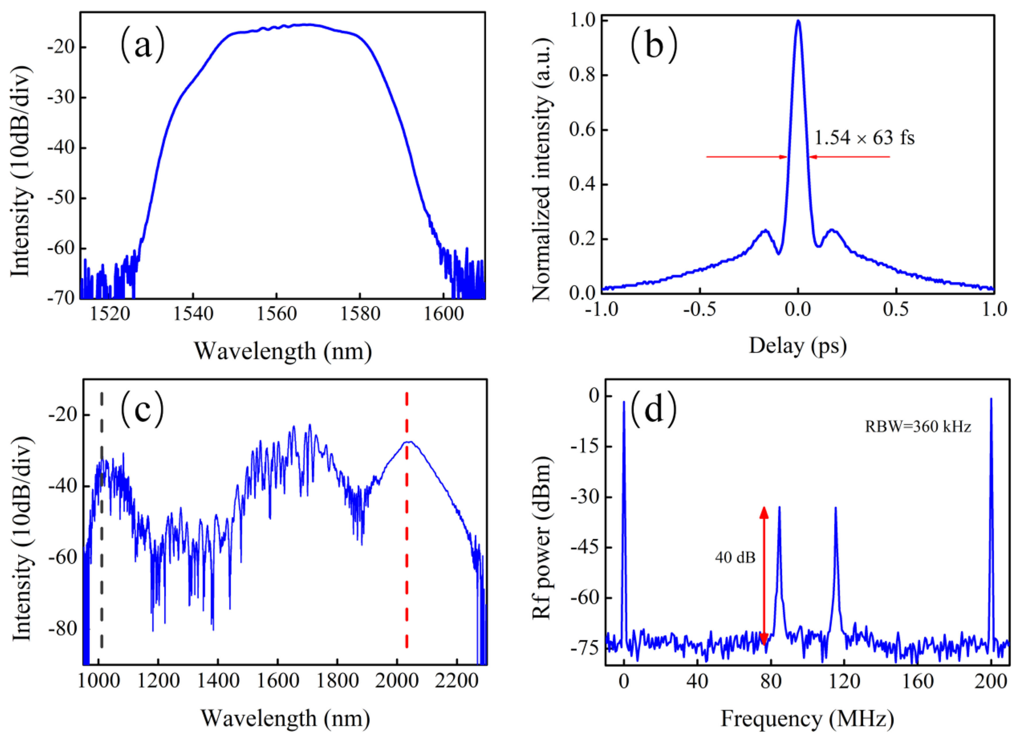

Fig. 3. (a) Output spectrum after a two-stage EDFA, (b) autocorrelation curve after pulse compression, (c) full octave-spanning SC spectrum after PM-HNLF, and (d) RF spectrum of the CEO beat signal.

inside the GRIN lens. Therefore, it is easy to optimize the pigtailed fiber length outside the GRIN lens. The fiber length is finally fine-tuned to 16 cm to achieve the temporal overlap between the two 1015-nm signals. The SC light coupled into the GRIN lens is collimated and focused in the middle of the PPLN crystal. After the frequency-doubling process, the generated light is collimated and focused again into the PM 1550 fiber through the second GRIN lens. The specially designed miniature f-to-2f interferometer structure not only reduces the size of the frequency comb, but also decreases the coupling loss. A CEO beat signal with an SNR of approximately 40 dB in a 360-kHz resolution bandwidth is detected using a PIN photodetector (Fig. 4(d)).

3. Static and Dynamic Response Characteristics of f_r

We used a PZT-driven, mechanical fiber stretcher with a large tuning range of f_r to stabilize the cavity length. The fiber actuator is a specially designed rigid-material-based unit that can be expanded by a high-voltage-driven PZT (Pantpiezo, PTJ1500505401, with a maximum displacement

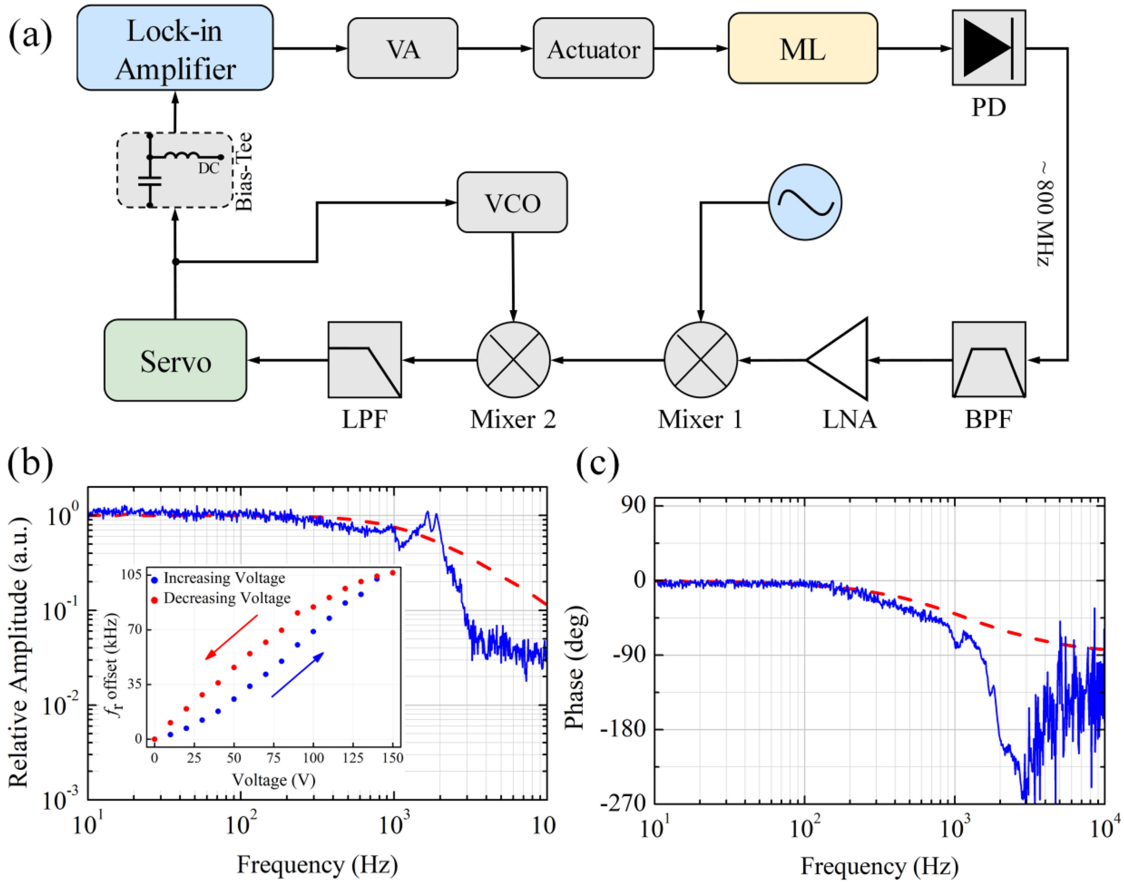

Fig. 4. (a) Schematic representation of transfer function measurement of f_r upon modulation. VA: Voltage amplifier, ML: mode-locked laser, BPF: bandpass filter, LNA: low-noise amplifier, LPF: low-pass filter, and VCO: voltage-controlled oscillator. (b) Relative amplitude of the transfer function of f_r for PZT modulation (blue) and VA output voltage for input voltage modulation (red); inset: static tuning curve of f_r. (c) Phase of the transfer function of f_r for PZT modulation (blue) and VA output voltage for input voltage modulation (red).

of 40 μm and capacitance of 2.4 μF). A 37-cm-long intracavity fiber is fixed on the fiber actuator with no damage. The mechanical fiber actuator and the oscillator are packaged in a compact steel box with a size of 128 $\times$ 81 $\times$ 9 mm^3.

The maximum tuning range of f_r is an important parameter to ensure the long-term stabilization of an OFC. To demonstrate the static tuning ability of the fiber actuator, the tuning curve of f_r as a function of the PZT voltage is measured. As shown in the inset of Fig. 4(b), the static curve for f_r exhibits hysteresis. This phenomenon is caused by the hysteresis source in PZT, crystalline polarization effects, and molecular friction. When the voltage applied to the PZT reaches 150 V, the tuning range of f_r increases to 106 kHz. Because the thermal expansion coefficient of fiber is 10 ppm/°C, the repetition rate tuning range of 106 kHz corresponds to an oscillator temperature change of ΔT $\approx$ 53 °C. Therefore, the tuning range is large enough to ensure long-term locking of f_r when the ambient temperature changes.

To characterize the dynamic response characteristics of the PZT-driven fiber actuator, the amplitude and phase response of f_r upon modulation were measured by using a frequency discriminator [27] and a lock-in amplifier (Zurich Instrument, UHF). The schematic of the dynamic response measurement system is shown in Fig. 4(a). More specifically, the voltage waveform at the input of the voltage amplifier (VA) (Piezo Drive, PDu150, with a gain of 20 V/V) was modulated by a sine signal output from the lock-in amplifier. Because the analog output of the lock-in amplifier is limited to $\pm$1.5 V, the voltage applied to the PZT was set to 18–30 V. The fourth harmonic of f_r at approximately 800 MHz was extracted by using a bandpass filter to increase the frequency change range after modulation. Then, the signal was amplified and mixed to the output frequency range of the voltage-controlled oscillator (VCO) by using a mixer (Mini-Circuits, ZAD-2+). The frequency down-converted signal was compared again in mixer 2 (Mini-Circuits, ZFM-2+) to the output signal from the VCO (Mini-Circuits, ZX95-200+, with a nominal tuning coefficient of 7 MHz/V in the 100–200 MHz range). The resulting error signal was low-pass filtered and fed into a servo to phase-lock the VCO to the output frequency of mixer 1. Then, the control signal of the VCO was used to reflect the frequency variation of the modulated f_r. By performing lock-in detection of the control signal of the VCO, the dynamic response of f_r in amplitude and phase upon modulation was attained. The lock-in detection was performed by the lock-in amplifier set to the applied modulation frequency. To protect the lock-in amplifier, a bias-Tee (Mini-Circuits, ZFBT-4R2GW+) was placed in front of the input end of the device to filter out the direct current component.

The measured amplitude and phase of the transfer function of f_r are shown in Figs. 4(b) and 4(c), respectively. The amplitude response behaves like a low-pass filter with a -3 dB cutoff frequency of approximately 1 kHz. A large resonance peak appears at a modulation frequency of approximately 1.65 kHz. For the phase response of f_r, a phase shift of approximately -81° occurs at 1 kHz, and a transient phase rebound occurs near 1.2 kHz, which is compatible with the amplitude attenuation. In addition, in the presence of the 2.4-μF capacitive load, the transfer function of the output voltage of the VA modulated by the same input voltage waveform from the lock-in amplifier is measured by performing the same lock-in detection. The observed response bandwidth is approximately 1 kHz in amplitude. Therefore, limited by the small signal response bandwidth of the VA, the response bandwidth of f_r is not less than 1 kHz. In summary, the main advantage of this fiber actuator is that it not only offers a large dynamic range to compensate the ambient temperature drift, but also has a high response speed to stabilize the repetition rate.

4. Phase Locking of f_r and f_{ceo}

The phase locking of the OFC is achieved by modulating the pump current of the oscillator and the fiber actuator. As shown in Fig. 5, for f_{ceo} stabilization, the 3 $f_r + f_{ceo}$ frequency signal (approximately 640 MHz) is bandpass filtered, amplified, and then frequency divided to approximately 80 MHz by using a frequency divider with a dividing ratio of eight. This signal is counted and compared with an 80-MHz reference frequency. The error signal output from the phase detector is low-pass filtered and then fed into the field-programmable gate array (FPGA) to lock the f_{ceo} by controlling the pump current of the oscillator. For f_r stabilization, the f_r at approximately 200 MHz is low-pass filtered, amplified, and mixed with a reference frequency of 200 MHz from a reference synthesizer. The resulting error signal is low-pass filtered and fed into an FPGA to produce the correction signal to regulate f_r. This correction signal is further amplified by the VA before application to the fiber actuator. To ensure long-term operation of the OFC, all synthesizers and counters are referenced to the same hydrogen maser.

Fig. 6(a) represents a 7200-s simultaneous stabilization record of f_r and f_{ceo} with two frequency counters (Agilent 53230A) at a gate time of 1 s. The Allan deviations of f_r and f_{ceo} with a 1-s gate time are 330 μHz and 63 mHz, corresponding to in-loop relative frequency stabilities of 1.65×10^{-12} and 7.9×10^{-10}, respectively. All frequency counters were referenced to the hydrogen maser.

To demonstrate the effect of the PZT-driven actuator on improving the noise property of the f_r signal, the frequency noise power spectral density (FN-PSD) of f_r in free-running and locked states were measured by using a signal source analyzer (Rohde & Schwarz, FSUP26). The measured FN-PSD of f_r is shown in Fig. 6(b). After locking, the noise level of f_r is significantly reduced at Fourier frequencies up to 1 kHz. The residual integrated timing jitter of the stabilized f_r signal is 418 fs [3 Hz to 1 MHz]. In addition, we found that multiple phase noise peaks near 1 to 2.5 kHz

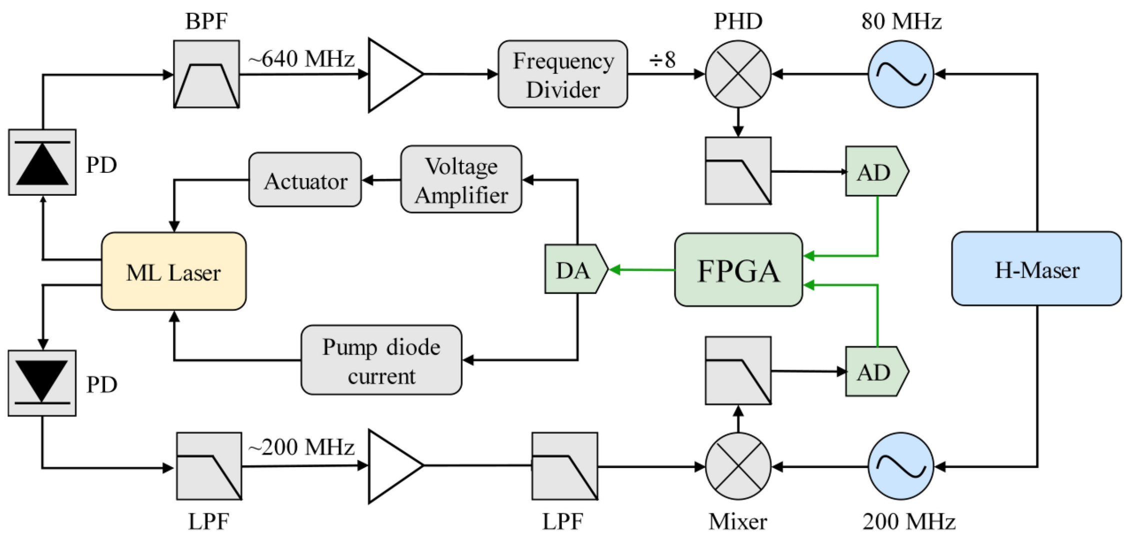

Fig. 5. Schematic of the locking electrics. PD: photodiode, PHD: phase detector, AD: analog-to-digital converter, DA: digital-to-analog converter, and FPGA: field-programmable gate array.

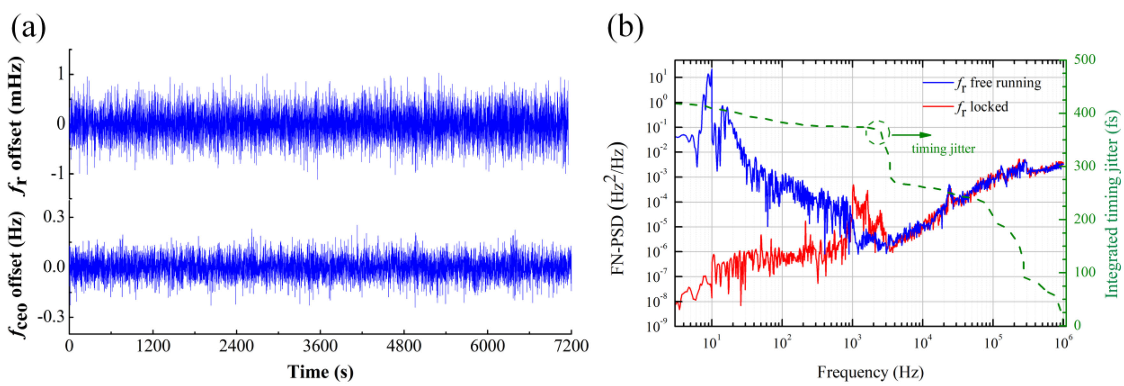

Fig. 6. (a) Counted frequency offset of f_{r} and f_{ceo} at a 1-s gate time while the OFC is fully stabilized. The Allan deviations of f_{r} and f_{ceo} for a measurement period of 7200 s are 330 μHz and 63 mHz, respectively. (b) Noise properties of f_{r}. Left axis: measured FN-PSD of the f_{r} in free-running (blue) and stabilized (red) state. Right axis: integrated timing jitter of the stabilized f_{r} signal as a function of the lower cutoff frequency.

occur in the f_{r} FN-PSD curve in the stabilized state. Obviously, resonance occurs at these Fourier frequencies. These results agree closely with the dynamic response of f_{r} for the PZT modulation.

5. Conclusion

We demonstrated a compact, robust OFC based on a 1.5-μm all-PM fiber mode-locked laser. A CEO beat with an SNR of 40 dB was attained by using a fiber coupled GRIN lens-based miniature f-to-$2f$ interferometer. In addition, we presented a PZT-driven fiber actuator with a large tuning range of 106 kHz and a high control bandwidth of approximately 1 kHz. By referencing a hydrogen maser, the in-loop relative frequency stabilities of f_{r} and f_{ceo} were found to be 1.65×10^{-12} and 7.9×10^{-10}, respectively, at a gate time of 1 s. In the phase-locked state, the frequency noise at

Fourier frequencies less than 1 kHz was significantly suppressed and the residual integrated timing jitter of f_r was 418 fs [3 Hz to 1 MHz]. Our design was effective in reducing the size of the OFC along with significantly decreasing the power consumption of the system. The performance of the OFC can be further improved by increasing the control bandwidth of the fiber actuator by using a PZT with a smaller displacement and capacitance.

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